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Jaime Aragones, Gustavo Torres, Francisco Torres, and
Catalina Segura | Supply Chain Management Case Study

Teaching Note

Sideraceros at a Crossroads: Scaling Beyond Tradition

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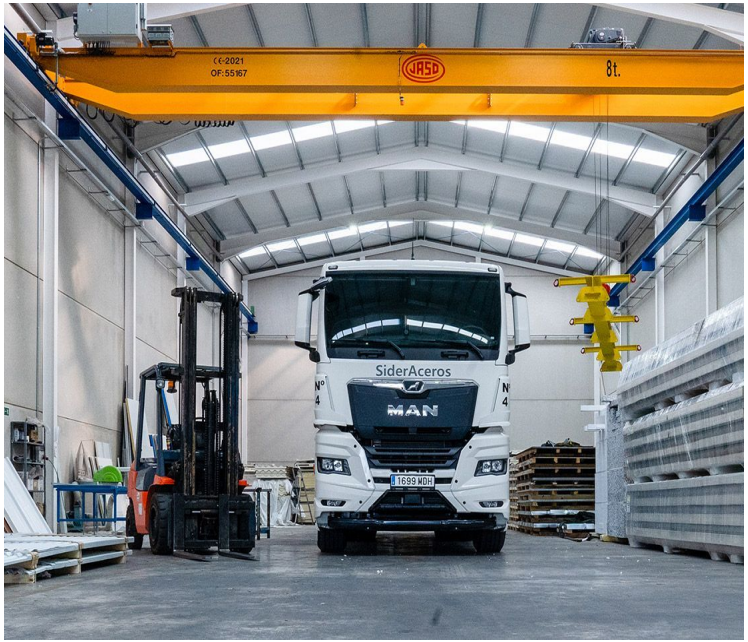
Case Summary

- Sideraceros, a **mid-sized steel distributor**, has grown significantly but continues to rely on outdated, manual route planning methods, leading to inefficiencies in deliveries and emergency handling.

- To address these challenges, the company hires Juan David, a young new hire tasked with i**dentifying process improvements and implementing a scalable solution for delivery optimization**.

- Students will analyze real-world logistics constraints, assess inefficiencies in manual routing, and apply optimization techniques to propose data-driven improvements.

- The case introduces key concepts in supply chain management, including vehicle routing problems (VRP), heuristic methods, and cost-benefit analysis in logistics decision-making



Task Overview

Assignment 1

How effective is Sideraceros' current route planning process? Identify inefficiencies and assess the risks associated with manual planning.

Purpose: Understand current company situation

Assignment 2

What was the total estimated cost of the delivery routes driven on 24/01, and can we manually design more efficient alternatives?

Purpose: Introduce topics and limitations of route optimization

Assignment 3

Can SiderAceros prepare for delays before they happen? Can routes be made to lower the risk of random events delaying delivery times?

Purpose: Provide a solution for emergency planning

Assignment 4

Does Sideraceros need additional vehicles to meet rising demand? How should they assess fleet capacity and utilization?

Purpose: Use utilization as a way of understanding fleet capacity

Assignment 5

What strategic steps should the company take to improve scalability? Based on the case information and previous questions what would be your final recommendation for the company?

Purpose: Analyze long-term strategy for decision making

Task Key Learnings

Assignment 1

Evaluating the current strategy

- Understand the limitations of traditional route planning.
- Recognize the impact of outdated processes on scalability and service quality.

Assignment 2

Route optimization problem

- Apply optimization techniques to improve delivery efficiency.
- Quantify potential savings in time, distance, and cost.

Assignment 3

Emergency prioritization

- Develop a systematic decision-making process for prioritization.
- Consider trade-offs between cost, time, and customer satisfaction.

Assignment 4

Vehicle utilization and fleet capacity

- Analyze whether current fleet size is sufficient for increased demand.
- Introduce methodologies for making data-driven fleet expansion decisions.

Assignment 5

Long term strategy

- Encourage strategic thinking around the long-term implications of inefficient processes.
- Develop a structured recommendation for implementing new systems.

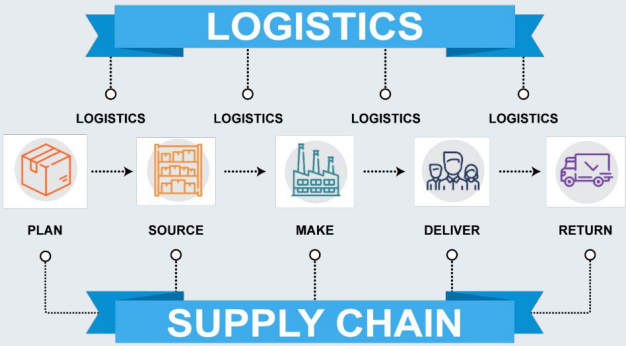
Time Frame for Solving

Assignment 1	Assignment 2	Assignment 3	Assignment 4	Assignment 5
Estimated time for reviewing the case and research: 15 min Estimated time for completing work: 15 min	Estimated time for familiarization of topics and research: 20 min Estimated time for completing work: 60 min	Estimated time for familiarization of topics and research: 20 min Estimated time for completing work: 40 min	Estimated time for familiarization of topics and research: 20 min Estimated time for completing work: 40 min	Estimated time for reviewing the case and research: 15 min Estimated time for completing work: 15 min

Final Output

The expected output is a slideshow presentation with the solutions for all assignments. For assignment 2 and 4 students can attach the work they did by hand to solve said problems as an appendix to the presentation.

Assignment 1



What is the most critical operational bottleneck currently affecting Sideraceros' delivery performance? Justify your answer by referencing specific examples from the case study.

Task to complete:

- Identify the operational bottleneck
- Why is it important to solve this first
- How this bottleneck impacts the business

A complete answer shows understanding the current planning process and a well though out matrix, framework or explanation for each of the sub questions.

Students must use information shared in the case study and annexes as well as previous knowledge they have on the topic.

Grading is based on each of the three questions answered completely.

Identifying the Operational Bottleneck: Manual Route Planning

5 Why's

- Problem: Delays, inefficiencies, high costs in deliveries

  → Why? Routes are inefficient and unpredictable

  → Why? They are manually planned with no optimization

  → Why? No route-planning software or historical data is used

  → Why? Company relies on Ana's experience

  → Why? No standardized or automated planning system exists

SiderAceros relies on Ana to manually plan delivery routes using paper-based methods and personal intuition. This creates a system that depends on a single person, does not allow for greater inflow of orders, and wastes more money than should be spent.

Manual route planning is the root bottleneck blocking efficiency, resilience, and scalability.

Why Manual Route Planning is the Bottleneck to Solve First

	Low Feasability	High Feasability
Low Impact	Improving paper records Better phone communication – Improves symptom, not root problem	Digitize delivery logs – Low effort, allows better analysis in future tasks
High Impact	Hiring more drivers - harder, expensive Buying more trucks - also hard and expensive	**Fix route planning system** - Improves software/process, reduces reliance on Ana

Prioritization Matrix: Impact vs. Feasibility

This matrix is used to understand why upgrading the system from manual planning makes the most sense for Sideraceros at this time. It also encourages students to think about other solutions to this issue.

How This Bottleneck Impacts the Business

Problems created by the bottleneck	Impact
Delivery delays	Unsatisfied clients, decrease in good reputation
Higher fuel/labor costs	Reduced margins, less money to invest in growth
No scale possible	No possibility to increase amount of clients and thus profit
No knowledge transfer	Over-reliance on one person creating extra risks if she is on vacation or quits
Delays in dispatching trucks	Idle labor hours (drivers waiting on Ana's decision)
No route consistency	Hard to train new drivers or transfer route knowledge
No performance tracking	Hard to evaluate or improve delivery KPIs

Assignment 2

What was the total estimated cost of the delivery routes driven on 24/01/25, and can we manually design more efficient alternatives?

Breakdown of tasks

- Calculate real delivery costs using provided locations
- Propose 2–3 alternative routes
- Reflect on manual planning vs optimization tools

Students should use the cost assumptions and location data provided which are based on real data Sideraceros delivery data in Madrid.

Grading should be based on completeness of all required tasks. There must be at least 2 new routes calculated along with an explanation if there are better or not than the one provided.

Truck delivery locations and actual route for 24/01/25

The actual route can be found under the QR Code or link found to the right. The current route for each truck is detailed in the table at Column A of the third sheet titled: “Appendix 1 for A2”.

When editing the sheet please create a copy.

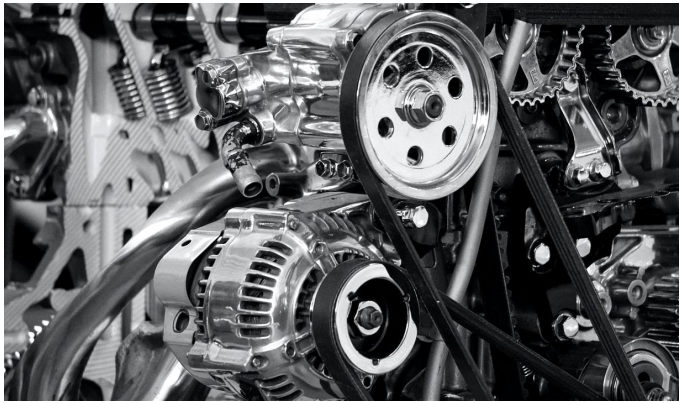
https://docs.google.com/spreadsheets/d/1tMVeaTJ4cQl-2rQfxwiPHAY-okthlAS3NTuM97qjfHk/edit?usp=sharing

Inputs and important costs

Item	Cost or Important Information
Truck Type	MAN TGM 26.340
Fuel Type	To be found by student
Fuel Consumption per Km	To be found by student
Avg. Maintenance Cost per Truck	€21.18/ day *
Km Driven by truck	To be found by student

* Maintenance cost is a set number based on data provided by Sideraceros. No further calculation or derivation needed.

Inputs and important costs



Item	Cost or Important Information
Engine Type	MAN TGM 26.340
Fuel Type	Diesel
Fuel Consumption per 100 Km	30 liters
Fuel Consumption per Km	0.30 L
Fuel cost per liter	1.53 EUR **
Fuel cost per km	0.459 EUR

** 3 month Madrid diesel price average (Source: Diesel prices in Madrid, Spain - 24-Mar-2025 | GlobalPetrolPrices.com. (2025). GlobalPetrolPrices.com. https://www.globalpetrolprices.com/Spain/Madrid/diesel_prices/)

Derivation of fuel consumption rate

Determining Fuel Consumption

- Average diesel truck consumption for every 100km driven is 30 to 40 liters
- MAN TGM 26.340 has **engine MAN D0836 LFL68 / Euro 6**
- If operating in mild climate a medium duty truck with engine MAN D0836 LFL68 engine consumes 23–33 liters per 100 km
- Average diesel truck consumption for every 100km driven is 30 to 40 liters
- Truck complies with Euro 6 emission standard
- Urban delivery (with stops and starts) uses more fuel than highway driving
- Heavier loads contribute to higher fuel consumption per km
- Given all of this information **we assume an average rate of 30 liters per 100 km** since Sideraceros trucks operates with a relatively efficient MAN engine and is used with varying loads in urban roads.

Sources

Bridgestone Mobility Solutions B.V, & Wise, B. (2024, November). Truck Diesel Use Per Mile | Fuel Calculator. Webfleet Blog; Webfleet. https://www.webfleet.com/en_gb/webfleet/blog/how-much-diesel-does-a-truck-use-per-mile/

MAN Truck & Bus SE. (2021). Www.man.eu. https://www.man.eu/global/en/truck/all-models/the-man-tgm-and-man-tgl/specifications/tgm_l-specifications.html

Which oil to use for your MAN TGM Euro 6 TGM 26.340 BB (6x4) Euro 6 (2012 and after)? A complete overview of all components such as the engine, transmission (transaxle), brake system, power steering, and cooling system. (2025). Kroon-Oil.com. https://www.kroon-oil.com/en/product-recommendation/trucks-and-buses-7-5t/man/tgm-euro-6/tgm-26-340-bb-6x4-euro-6/46386/

Total cost to operate all trucks for 24/01

Truck	Total Distance (km)	Fuel Cost (EUR)	Maintenance Cost (EUR)	Total Cost for 24/01 (EUR)
4741JYY	200.5	92.03	21.18	113.21
1699MDH	125.7	57.70	21.18	78.88
2275KYK	152.2	69.86	21.18	91.04

Total for all 3 trucks: 283.13 EUR

Total Distance by Truck

Distances were determined by inputting delivery coordinates into Google Maps and using the first recommended driving route.

Methods for designing better routes

Manual route optimization methods

- **Nearest Neighbor per Truck:** Assign each truck its own group of stops and visit the closest next location until all are completed.
- **Clustering + Route Planning:** Group delivery points into 3 geographic clusters and assign one truck to each cluster for simpler, non-overlapping routes.
- **Round-Robin Assignment:** Distribute delivery points one by one to each truck in order, then optimize each truck's path separately.
- **Route Balancing:** Adjust routes to ensure each truck has a similar total distance or number of stops to balance workload and time on the road.
- **Trial & Error with Cost Comparison:** Try different truck-location assignments manually, calculate cost for each (fuel + maintenance), and choose the lowest-cost option.

Constraints for designing new routes

Clarification of how new routes can be built for solution standardization

- New routes can be created with a minimum of 1 truck and a maximum of 3 trucks.
- Each truck can cover a maximum of 250km in one day.
- For simplicity the truck has to visit the Sideraceros warehouse at least 2x in one day, the first does not have to be in any particular order or position but they have to finish at Sideraceros
- You are allowed to change what stop is assigned to what truck, not consolidate the number of visits.
    - i.e. if truck 1 visit Location A and truck 2 also visits Location A, you are allowed to make the change so that truck 1 visits Location A twice but at different points in the day.

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Proposed alternative route

The new route can be found under the QR Code or link found to the right. The new route for each truck is detailed in the table at Column M of the third sheet titled: “Appendix 1 for A2 w/ Solution”

When editing the sheet please create a copy.

https://docs.google.com/spreadsheets/d/1tMVeaTJ4cQI-2rQfxwiPHAY-okthlAS3NTuM97qjfHk/edit?usp=sharing

Optimized routes create cost savings

Comparison of results

Truck	Total Distance (km)	Fuel Cost (EUR)	Maintenance Cost (EUR)	Total Cost for 24/01 (EUR)
4741JYY	163.1	74.86	21.18	96.04
1699MDH	119.5	54.79	21.18	75.97
2275KYK	137.6	63.19	21.18	84.37

Total for all 3 trucks: 256.39 EUR

Reflection and key takeaways

- Significant cost savings even with **manually optimized route reduced costs from €283.13 to €256.39** and when scaled across multiple trucks and days, this leads to substantial savings.
- Manual planning is inefficient: This exercise highlights how time-consuming and error-prone manual route planning can be.
- Scalability is a challenge, as locations and trucks increase**, route permutations grow exponentially**, making manual planning unsustainable.
- The **need for route optimization software** becomes clear. These tools streamline planning, reduce costs, and support smart decisions.

Assignment 3

Given SiderAceros's current manual route planning system and the frequent mid-day emergencies handled by Ana, how can the company reduce delivery delays and create more resilient delivery routes while maintaining customer satisfaction?

Tasks to complete:

- Analyze Current Delivery Data
- Classify Delay Types
- Design a Mitigation Strategy

Expected Output:

- Delay Analysis Summary
- Preventability Matrix
- Risk Mitigation Framework

Students should use the available delivery data, knowledge of supply chain inter-networks, and intuition of possible delay contributor (condensed metro areas, geographic characteristics, weather, mechanical, etc.) in order to output a guide that allows for a systematic removal of delays. The information is provided in the next few slides.

Delivery Information

Our delivery is scheduled to be completed on March 7th at 3pm to a customer who is located in Madrid Centro. We must find a way to create a delivery plan (using the stated tasks) that minimizes our delays.

March	Friday
Madrid Centro	3pm

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Current delivery data

Based on the current delivery data, we can start looking at certain times that are more prone to delays. This gives us a good idea of when we have to plan the most in order to avoid delays.

1. **Early Morning (7:00 - 9:00)**
   - Delivery times are relatively fast compared to the rest of the day.
   - Example: Many deliveries have a duration of 15-30 minutes.
2. **Late Morning (9:00 - 12:00)**
   - More deliveries with slightly longer times start to appear.
   - Some orders start accumulating delays.
3. **Noon and Early Afternoon (12:00 - 15:00)**
   - Several deliveries exceed 45 minutes in duration.
   - Breaks (e.g., from 14:00 to 15:00) may be affecting delivery speeds.
4. **Afternoon (15:00 - 17:00)**
   - This is when the most significant delays seem to occur, with several deliveries taking over an hour.
   - Deliveries in this time frame appear to be the most affected by delays.

Assignment 3

Classifying Delay Types

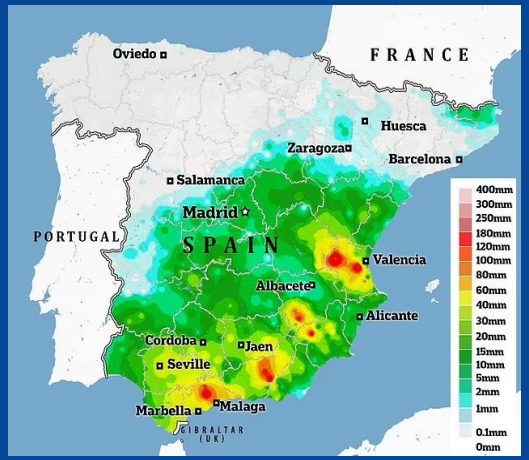
Many times certain delays are attributed to chance but, in most case they are caused by under planning or assuming ideal conditions. In reality if we assume the worst condition possible and plan around it we will have the best performance possible regardless of good or bad conditions.

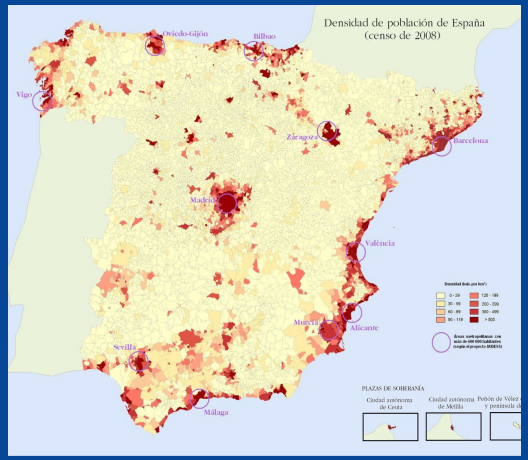
Delay	Prevention	Predictability
Mechanical	*Timely Repairs*	*High*
Weather	*Planning around forecasts*	*Variable*
Traffic	*Avoiding densely populated areas during rush-hour*	*Moderate*
Unprepared Material	*Better communication between teams*	*High*
Planned breaks	*Strategically plan breaks to avoid events that may delay deliveries*	*High*

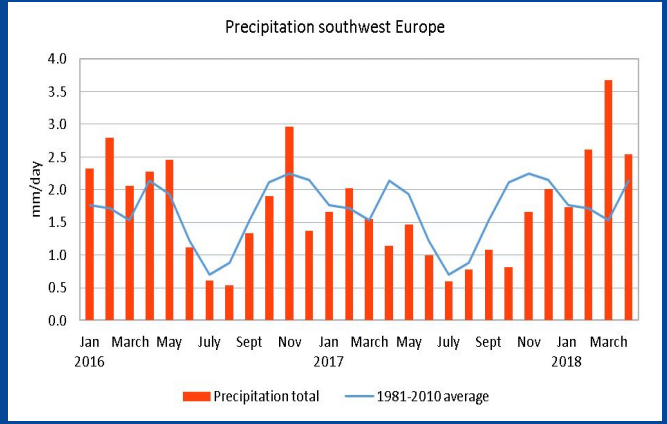
Assignment 3

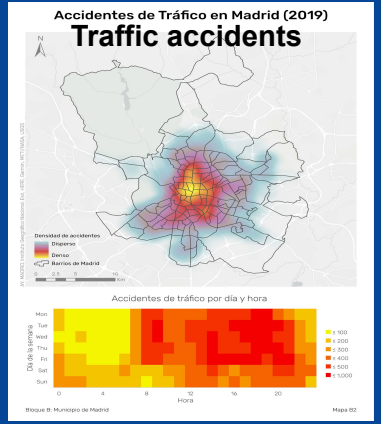
Visualization of geographical data , weather, population density, etc

By analyzing data such as areas prone to floods (given Spain's recent floodings), average rainfall per month and region, areas with high population density, and traffic patterns, we can foresee delays due to floods and traffic as well as what months and regions are more susceptible to such events.









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Final results: Input

Analysis of Current Delivery Information

1. **Early Morning (7:00 - 9:00)**
   - Delivery times are relatively fast compared to the rest of the day.
   - Example: Many deliveries have a duration of 15-30 minutes.
2. **Late Morning (9:00 - 12:00)**
   - More deliveries with slightly longer times start to appear.
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3. **Noon and Early Afternoon (12:00 - 15:00)**
   - Several deliveries exceed 45 minutes in duration.
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4. **Afternoon (15:00 - 17:00)**
   - This is when the most significant delays seem to occur, with several deliveries taking over an hour.
   - Deliveries in this time frame appear to be the most affected by delays.

Delay trends and patterns

Framework for systematic removal

Categorization of Delays

Delay	Prevention	Predictability
Mechanical	*Timely Repairs*	*High*
Weather	*Planning around forecasts*	*Variable*
Traffic	*Avoiding densely populated areas during rush-hour*	*Moderate*
Unprepared Material	*Better communication between teams*	*High*
Planned breaks	*Strategically plan breaks to avoid events that may delay deliveries*	*High*

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Final results: Output

Example: Framework for Systematic removal

Delay	Available Information	Prevention
Mechanical	Trucks have been running for 3 weeks without rest and haven't had check-ups in 4 months	*Have rigorous maintenance on trucks every 2 months, general condition checks every 2 weeks and daily checks for oil levels*
Weather	It is March 7th and we expect the highest level of rain of the year	*Rain usually starts after 11, plan deliveries to floodable, or high rainfall areas early in the day*
Traffic	Our delivery is planned for Friday 3pm in Madrid Centro, high traffic expected, high probability of accidents	*If delivery cannot be completed during the morning attempt to deliver it during the 10-12 reduced traffic*
Unprepared Material	The last 6 deliveries to this customer have had 15 minutes of delays due to unorganized client	*Inform Client of arrival time and give a 10 minute arrival window so that they have time to prepare*
Planned breaks	1 hour break is planned at 11am	*Move break to a high traffic time such as 12-13 or 14-15 in order to maximize conditions*

Output

Based on the information collected we can make an adjusted plan for our delivery scheduled for Madrid Centro on Friday the 7th of March. Our delivery will be made during the early morning 9-11 in order to avoid traffic and rain, our customer will be informed of our arrival in a timely manner, and our drivers break will be from 12-13 so that he gets a chance to get out of the high congestion area and pass the rush hour traffic stationary.

Analysis of final results

Final results

Current delay Analysis

Framework to systematically reduce risk

Categorization of delays and solutions to it

Grading

Based on student's ability to demonstrate that he/she can conduct an analysis of delay-contributing factors, understand the causes behind them and search for methods to mitigate such delays

Reflection and key takeaways

Current delay analysis: clear pattern of delivery delays throughout the day (15-17 highest frequency of delays {people leaving work, or picking up children})

Categorization of delays and solutions: labeling delays as highly predictable can help the planner (human or computer) focus on specific planning to avoid such delays and to have broader plans for less predictable events

Framework to systematically reduce risk: Once all potential risk factors are analyzed and categorized, we look up information that is directly correlated to these risks and come up with a solution

Assignment 4

Does Sideraceros need additional vehicles to meet rising demand? How should they assess fleet capacity and utilization rate?

Students will estimate vehicle utilization based on delivery data and use this to decide whether additional capacity is needed.

Breakdown of tasks

- Calculate utilization per vehicle based on data provided
- Analyse results
- Give a final recommendation

Expected output are slides showing calculating utilization per vehicle and a final analysis of results and final recommendation.

Grading will be based on completeness of all requirements of the task.

Utilization as a defining metric

- Definition: Utilization (ρ) is a measure of how intensively a resource (truck, server, or machine) is being used. It is calculated as the ratio of time spent working to total time available.
- In this case, utilization is based on hours spent on deliveries compared to the total shift time per truck per day.
- Valid Range: Utilization values must lie between 0 and 1.
  - $\rho \approx 0 \rightarrow$ underutilized (excess idle time)
  - $\rho \approx 1 \rightarrow$ operating at or near full capacity
- Formula: ϱ = Average Time Spent Working/ Total Available Time
- Implications of High Utilization: As ρ approaches 1, queues and delays increase sharply, particularly in stochastic systems where demand and service times vary. This signals a risk of system breakdown if demand spikes or disruptions occur.
- Application in Case: With the 20% projected increase in deliveries from the new client, current utilization levels should be re-evaluated. If any truck's ρ nears or exceeds 1 after adjustment, this may justify adding a new truck or rebalancing routes to avoid overloading the system.

Inputs and required data

Data for 12/12

Truck	Start Time	End Time	Number of Stops	Avg. Stop Duration (mins)	Average company work day (hours)
Red	8:02	16:28	8	42	9
Green	7:16	16:32	8	33	9
Blue	7:44	16:38	10	35	9

Assignment 4

Calculating Utilization

Data for 12/12

Truck	Start Time	End Time	Average company work day	Utilization
Red	8:02	16:28	9 hours	0.94
Green	7:16	16:32	9 hours	1.04
Blue	7:44	16:38	9 hours	0.98

Accounting for 20% increase in demand

Breakdown of Impact

- Estimate Current Number of Stops per Driver:
  - Red (Abraham): 8 stops
  - Blue (Rubén): 9 stops
  - Green (Jefferson): 10 stops
  - Total: 27 stops
- Apply a 20% Increase to Total Stops:
  - New Total Stops = 27 × 1.2 = 32.4 ≈ 32
- Estimate Average Time per Stop per Driver:
  - Red (Abraham): 8 stops
  - Blue (Rubén): 9 stops
  - Green (Jefferson): 10 stops
  - Avg: 27 stops
- Estimate Time Needed for 5 More Stops (to go from 27 → 32):
  - 5 stops × 36 min = 180 min = 3 hours extra
  - This does not even include distance to get there

Final Results

- The current 3 trucks are already working near or above full capacity:
  - Abraham: 0.94 utilization
  - Rubén: 0.98 utilization
  - Jefferson: 1.04 utilization
- Adding 5 more stops (~5 extra hours) would exceed daily work limits.
-
- The company needs to consider:
  - Adding a 4th truck
  - Designing better routes with optimization software to make them more efficient
  - Using overtime (has extra costs and reduced reliability)

Assignment 5

Sideraceros is at a turning point, facing pressure to modernize its delivery operations while scaling its business efficiently. Based on the case study and previous assignments, develop a strategic plan that balances operational efficiency, cost-effectiveness, and organizational readiness for change.

Breakdown of tasks

- Business process reengineering (BPR)**:** Redesigning the Delivery Planning Process
- Cost benefit analysis: Justify the investment
- Change management strategy: Implementing and Sustaining the Transformation

Grading should be based on the proposals given for scalability in the company and if they have a way to instill a culture of seeking innovation in its employees.

A complete answer shows understanding of the concepts introduced in previous assignments and knowing which to introduce based on the company's current situation and pain points.

Scalability is limited by pain points that are affecting the company

Pain Points

- Delivery routes are planned on paper
- Lack of historical data
- Emergencies and unexpected changes disrupt operations
- Absence of digital tools
- Limited knowledge on innovative practices

Scalability as a strategic objective

- Ability to handle more orders with the same or less resources
- Adaptability to crises without interfering with the system
- Performance consistency as client base grows

Assignment 5

Apply business process reengineering (BPR)

Goal

Redesign the delivery planning process from scratch.

Steps

Map current workflow
(manual routing, daily logs, emergency handling)

↓

Identify inefficiencies
(no digitization, reactive not proactive)

Propose a new process

Centralized route planning software

↓

Real-time tracking and response system

↓

Use of historical data for forecasting

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Conduct a cost benefit analysis

Estimated potential savings

- Reduced fuel costs through optimized routes
- Less overtime from drivers
- Better customer satisfaction could lead to more contracts

Compare against costs

- Investment in routing software
- Training time for employees
- Temporary adjustment costs

Example: If better routing saves 100 km per day at €0.51/km = €51/day = €18615/year per truck.

Introduce a change management framework

Part 1

Build urgency

("Client demands are rising")

Part 2

Establish a guiding body inside the company

(Juan David + management sponsor)

Part 3

Communicate the vision to the employees

("Tech-enabled growth")

Part 4

Provide short-term accomplishments and implement the change in the company's culture

(show the improvements from data digitization and train staff)

Conclusion

Address the resistance to change that the company has (culture of long-standing employees) and implement Kotter's 8 Steps or ADKAR Model